DianGPT: Bridging Precision and Efficiency for a Domain-Specific Question-Answering System

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***Abstract*—In this paper, we present DianGPT, a large language model (LLM) tailored for specialized question-answering tasks through integration with organizational archives and domain-specific knowledge bases of the Dian Group. To address the limitations of existing models in balancing precision with resource efficiency, the proposed system combines parameter-efficient Low-Rank Adaptation (LoRA), Retrieval-Augmented Generation (RAG), and structured data analysis. Our approach ensures the model dynamically retrieves contextual information while maintaining computational efficiency. Our experimental results demonstrate that DianGPT, when augmented with RAG, achieves marked improvements in response quality and task-specific performance, validated through rigorous benchmarking. The proposed framework also emphasizes evaluation transparency, prioritizing traceable metrics to assess reliability. By advancing methodologies for domain-specific language models, we explore practical engineering techniques for developing intelligent Q&A systems in specialized fields while offering scalable insights for adapting LLM in diverse applications. As a case study, DianGPT demonstrates the potential of hybrid fine-tuning and retrieval strategies to balance performance with resource efficiency in real-world deployments.**

***Index Terms*—Large Language Model, Domain-specific Q&A, Low-Rank Adaptation, Retrieval-Augmented Generation**

I. Introduction

Founded in 2002 as a pioneering student group at Huazhong University of Science and Technology, the Dian team has fostered top engineering students consistently driven by technological innovation and knowledge management. As the team expanded, its growing influence and frequent cross-cultural exchanges, particularly in hosting interested visitors, revealed a critical group need: optimizing information access while preserving institutional knowledge. To address this, we developed DianGPT, a domain-specific intelligent question-answering system leveraging large language models (LLMs) to enhance operational efficiency and demonstrate LLM-driven innovation for academic organizations.

Modern LLMs have enabled widespread adoption of general-purpose Q&A systems, yet these solutions struggle with specialized domains. Key limitations include insufficient domain knowledge integration, generic responses lacking contextual relevance, and limited customization options [1]. For organizations like the Dian group, which rely on decades of curated records (e.g., meeting minutes, yearbooks, and project archives), existing systems fail to balance accuracy with computational efficiency.

Traditional approaches using generic training data further compound these challenges [2]. By contrast, DianGPT introduces a tailored architecture that systematically incorporates the group's institutional knowledge through LLM fine-tuning. Our system prioritizes two design objectives:

1) Domain-specific accuracy: Enhancing response relevance by integrating historical records while minimizing computational overhead.
2) Evaluation reliability: Establishing robust mechanisms to assess answer quality during iterative improvements.

DianGPT contributes to bridging the gap between foundational LLM capabilities and practical domain-specific requirements, offering a replicable framework for academic and industrial teams seeking to transform archival knowledge into actionable LLM applications. We design a domain-specific QA prototype system which addresses the limitations of general-purpose language models by integrating specialized knowledge, yet faces challenges in corpus integration and model adaptation. We also propose an automated evaluation framework to improve objectivity but struggle to balance efficiency with domain-specific accuracy including two core components. LoRA mitigates computational bottlenecks by enabling lightweight fine-tuning, preserving foundational model capabilities while tailoring outputs to niche tasks. RAG complements these efforts by dynamically retrieving domain-specific external knowledge during inference, enhancing factual precision. Together, these advancements in specialized adaptation, efficient evaluation, and knowledge-aware generation form an integrated application framework for optimizing precision-efficiency trade-offs in domain-specific QA systems.

II. Related Work

In this section, we present the recent research progress of the proposed integrated intelligent Q&A framework in the following 4 aspects.

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A. Domain-Specific Intelligent QA Systems

While LLMs have revolutionized general-purpose QA systems, their performance in specialized domains remains limited due to insufficient integration of domain-specific knowledge [3]. To address this gap, researchers have developed QA systems tailored to niche fields by leveraging domain-adapted training data and curated corpora. However, efficiently processing and harmonizing these corpora presents a significant challenge, as technical jargon and structural variations across domains complicate automated knowledge extraction [4]. Traditional approaches, such as manual knowledge base construction, scale poorly with increasing domain complexity, prompting a shift toward methods that combine LLM's generative capabilities with targeted corpus filtering.

A parallel challenge lies in adapting general-purpose pre-trained models (e.g., GPT, BERT) to domain-specific tasks. Although techniques like fine-tuning, domain adaptation, and transfer learning are widely used to enhance relevance [5], their effectiveness hinges on balancing model generalization with domain specialization. It remains an open problem to select optimal adaptation strategies, particularly for precision-critical applications, as overly narrow customization risks undermining the broad reasoning strengths of foundational LLMs.

B. Automated QA Evaluation

The evaluation of domain-specific QA systems has evolved from reliance on human-centric scoring to increasingly automated methodologies. Traditional approaches, where human evaluators rate answers based on criteria such as similarity and accuracy, are resource-intensive, subjective, and difficult to scale. These limitations often compromise objective performance assessment.

Automated evaluation frameworks leveraging metrics such as BLEU [6] and ROUGE [7] have gained prominence for their ability to quantify text quality. However, such general-purpose metrics struggle to capture domain-specific accuracy [8], as they prioritize lexical overlap over task-specific relevance. Recent work has thus introduced specialized evaluation metrics tailored to niche domains. For instance, QA accuracy [9] compares system outputs against expert-curated reference answers to assess their quality in a specific domain. Hybrid approaches further refine objectivity by combining automated scoring with human oversight. For example, large language models (e.g., GPT-4) are employed as automated evaluators to rate system outputs consistently [10].

Despite these advancements, critical challenges persist. Multimodal evaluation remains underdeveloped, particularly for cross-domain or highly customized tasks, as current frameworks lack adaptability to diverse data types and domain-specific requirements [11]. Future evaluation systems may need to integrate multidimensional feedback (e.g. user experience and iterative human input) to enhance reliability and reflect real-world utility.

C. LoRA Fine-Tuning

The rapid growth of LLMs has escalated the computational costs and time required for training and fine-tuning, posing challenges for domain-specific adaptation under resource constraints. LoRA has emerged as a prominent method for efficient fine-tuning. By introducing low-rank matrix updates rather than modifying all model parameters, LoRA achieves performance comparable to full-parameter optimization at a fraction of the computational cost [12]. This approach preserves the foundational knowledge of pre-trained LLMs while enabling targeted adaptation to specialized tasks [13].

LoRA's efficiency is particularly advantageous for domain-specific QA systems, where resource limitations often hinder the deployment of fully retrained models. By balancing generalization and specialization, LoRA reduces the overhead of adapting LLMs to niche domains without sacrificing accuracy. Beyond efficiency gains, LoRA also establishes a scalable framework for developing customized QA systems, demonstrating its versatility across diverse applications. Looking ahead, LoRA's lightweight adaptation paradigm holds significant potential for extending LLM capabilities to larger-scale and more complex domain-specific tasks, where traditional fine-tuning methods remain prohibitively expensive.

D. Retrieval-Augmented Generation

RAG addresses a critical limitation of conventional generative language models in that their reliance on static, parameter-embedded knowledge. This limitation often hinders performance on tasks requiring specialized or dynamic information [14]. By integrating real-time retrieval from external knowledge bases during inference, RAG enriches generated outputs with contextually relevant data, enhancing both accuracy and factual grounding.

Recent research has prioritized optimizing RAG's dual components, including both retrieval and generation, to better serve domain-specific applications. For example, dense retrieval methods, which prioritize semantic similarity over keyword matching, have gained prominence for their ability to efficiently identify contextually pertinent documents [15]. Further advancements involve joint training of the retriever and generator, enabling end-to-end adaptation to specialized domains such as healthcare and law. This synergy ensures retrieved information aligns closely with the generator's task objectives, improving coherence and domain relevance [16]. These innovations demonstrate RAG's potential to bridge the gap between broad language understanding and precision in niche applications, which provides the foundation critical to advancing systems like DianGPT in resource-constrained environments.

III. Methodology

As shown in Fig. 1, our system design mainly consists of 3 major steps including dataset generation, model training, and model inference. We also discuss key modules such as model evaluation and RAG in details in this section.

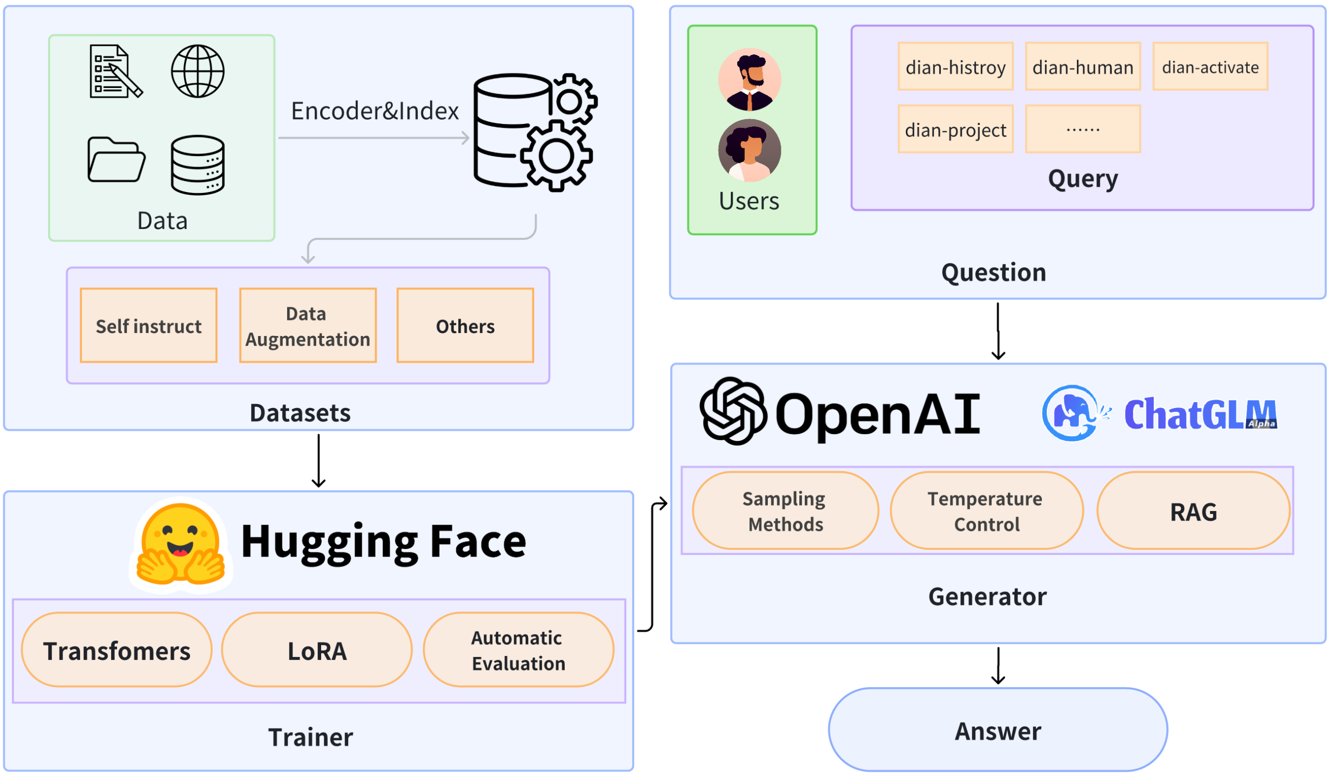

Fig. 1. Overview of the DianGPT system architecture

A. Dataset Generation

DianGPT employs structured knowledge extraction and standardized generation to construct a high-quality dataset for model training. The process involved the following key steps:

1) Raw Data Processing and Filtering: The almanac text, our primary data source, contained noise such as charts, photos, and sensitive personal data. We applied a section-based processing approach, segmenting the text by chapters as independent data units. Non-text contents were removed, sensitive data were redacted, and disorganized sections were consolidated to maintain thematic consistency.

2) Entity Extraction and Relationship Analysis: We utilized large language model APIs for entity extraction, focusing on organizations, events, individuals, and their relationships [17]. To mitigate referential ambiguity, we designed an enhanced prompt guiding the model to analyze entities within their relational contexts. This approach improved entity accuracy and ensured clear articulation of relationships, enhancing the dataset's quality.

3) Data Normalization and Generation: To ensure high-quality training data, we adopted a multi-turn dialogue generation approach, balancing completeness, diversity, and complexity. Normalization tools standardized data formats, while a data format checker validated structural consistency, coherence, and informational integrity.

4) Data Deduplication and Quality Control: Minhash-based deduplication removed redundant data, preserving the dataset's diversity and reducing overfitting risks [18]. This ensured the independence and richness of the training data.

5) Final Dataset Construction: The final dataset integrated structured, entity-extracted, normalized, and deduplicated data, covering historical events, individual backgrounds, and organizational structures of the Dian group. This dataset provided accurate, domain-rich knowledge while minimizing irrelevant content, improving model accuracy and usability.

B. Supervised Fine-Tuning

Supervised Fine-Tuning (SFT) was employed to adapt the general pre-trained model GLM4-9B for domain-specific question-answering tasks related to the Dian group's almanac and knowledge base. We leveraged LoRA to enhance efficiency while maintaining high performance.

1) Base Model and Fine-Tuning Objectives: GLM4-9B was chosen for its strong language understanding and generation capabilities, particularly in Chinese [19]. Its 9B parameter size strikes a balance between computational efficiency and performance, making it well-suited for domain-specific adaptation. Rather than training from the scratch, we fine-tuned the model to specialize in the Dian group's knowledge and Q&A tasks.

2) LoRA Fine-Tuning: To reduce computational cost and storage demands, we applied LoRA, which fine-tunes only a subset of model parameters using low-rank matrices. LoRA introduces additional low-rank weight matrices to optimize specific layers while preserving most of the base model's pre-trained knowledge. This approach allowed us to efficiently adapt GLM4-9B to the almanac data while enhancing accuracy and response quality.

3) Fine-Tuning Process and Strategy: We utilized the Adam optimizer with a Cosine learning rate scheduler for stable training [20]. The Cosine scheduler provided an initially high learning rate for faster convergence, which gradually decreased to improve generalization. Additionally, a Warmup strategy was applied, allocating the first 1% of the training steps to gradually increase the learning rate, mitigating issues such as the gradient explosion.

4) Model Merging: After fine-tuning, the LoRA-generated low-rank matrices were merged with the original model's weights, producing the final fine-tuned model. This version integrates both general pre-trained knowledge and domain-specific expertise, enabling high-quality Q&A generation.

C. Retrieval-Augmented Generation

To further enhance the model's generation capabilities for domain-specific tasks, we adopted the Retrieval-Augmented Generation (RAG) method. RAG combines external knowledge retrieval with the generative model, making the generated results more accurate and information-rich. The implementation of RAG in this study includes the following key steps:

1) Knowledge Index Construction: We used BCEmbedding [21] to efficiently embed the almanac text from the knowledge base, capturing the semantic features of the text. The embedded representations are stored in the FAISS (Facebook AI Similarity Search) retrieval engine, which serves as an external knowledge source for the model when generating answers [22].

2) Retrieval and Ranking: In the generation task, based on the user's input query, the model vectorizes the query and retrieves the most relevant candidate texts (top-3 results) from the FAISS library. To further improve the accuracy of the retrieval results, we introduced a re-ranking model to optimize the ranking of the initial retrieval results, ensuring that the content added to the generation context is highly relevant to the user's query.

3) Context-Augmented Generation: The top-3 retrieved text fragments are used as the context information and directly concatenated into the model's input prompt, providing richer domain knowledge support for the generation task. By integrating the retrieval results with the user query, the model generates logically coherent and semantically accurate answers.

Through the RAG approach, the model is not only able to utilize its built-in pre-trained knowledge, but also dynamically access externally retrieved domain-specific knowledge. This significantly improves the model's performance in vertical domain question-answering tasks. This method effectively mitigates the potential knowledge gaps that large models may have in smaller, specialized domains.

D. Automatic Evaluation

To effectively assess DianGPT's performance and provide timely feedback for the model optimization, we developed

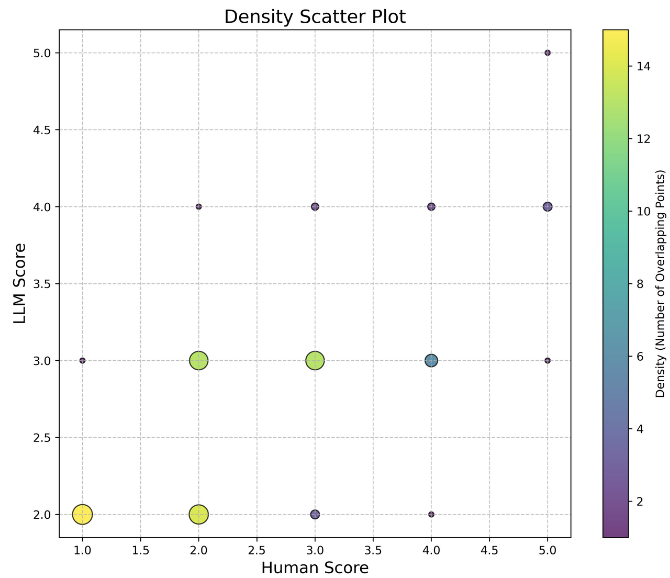

Fig. 2. Scatter plot showing the correlation between the marks scored by the LLM and the human evaluators

a quantitative evaluation framework. This framework aims to break through the limitations of traditional sample-by-sample observation, enabling a comprehensive and systematic evaluation of the model's performance across various tasks and question categories.

1) Design Objectives:

1) Quantitative Evaluation: Using a quantitative scoring system, the model's performance on different questions is comprehensively assessed, avoiding the reliance on manual sample-by-sample checks.
2) Quick Feedback: The framework provides efficient and timely feedback for model optimization, helping the development team identify the model's strengths and weaknesses, and allowing targeted improvements.
3) In American English, commas, semicolons, periods, question and exclamation marks are located within quotation marks only when a complete thought or name is cited, such as a title or full quotation. When quotation marks are used, instead of a bold or italic typeface, to highlight a word or phrase, punctuation should appear outside of the quotation marks. A parenthetical phrase or statement at the end of a sentence is punctuated outside of the closing parenthesis (like this). (A parenthetical sentence is punctuated within the parentheses.)

2) Evaluation Dataset and Scoring Criteria: Our evaluation used a custom-built DianGroupEval dataset, which contains 160 test samples covering different types of questions. These samples consist of high-quality, human-curated single-turn question-answer pairs, covering topics such as Dian Group's member information, history, characteristics, rules, and regulations. Each sample includes a question, a reference answer, and an expert score. The dataset is structured as follows:

- **Dataset Size**: The dataset consists of 160 question-answer pairs.
- **Dataset Structure**: Each entry in the dataset is structured with three components: a question, a reference answer, and an expert score.
- **Open Source**: The data curation process of the DianGPT training dateset is presented in this paper. The dataset used in this study can is publically available [23].

The scoring criteria adopted a 5-point scale, as follows:

- 1 point: Completely inaccurate, the response is severely incorrect and has no reference value.
- 2 points: Partially accurate, includes some correct information but contains obvious errors or significant omissions, misleading.
- 3 points: Largely accurate, the information is generally correct, but contains minor errors, omissions, or lacks depth in key content.
- 4 points: Highly accurate, the information is correct and comprehensive, but slightly lacking in logical presentation, conciseness, or fluency.
- 5 points: Completely accurate, the information is correct and comprehensive, with clear expression and rigorous logic.

3) Evaluation Process : First, the model to be tested is used to infer all the questions in the DianGroupEval dataset, and the model's outputs are saved for subsequent analysis. Next, a strong Chinese general-purpose model is selected as the judge model, and customized prompts are designed so that the judge model can score all responses from the tested model based on the sample's Query (question), Response (model answer), and Reference (reference answer). Finally, the average score of all response samples is used as the judge model's final evaluation result for the tested model. Additionally, Rouge-L F1 Score was utilized as a supplementary evaluation metric to objectively evaluate the semantic and structural similarity between the model's outputs and reference answers.

4) Evaluation Reliability: To ensure the reliability of the evaluation, we calculate the correlation coefficient between the human scores and the judge model's scores to measure the consistency of the evaluations. Specifically, a linear regression fitting method is used to calculate the correlation coefficient between the human scores and the judge model scores for 61 samples. An ideal correlation coefficient should be 0.6 or higher, indicating a high correlation between the judge model's evaluation and human scoring. If the correlation coefficient is below this value, we will adjust the choice of the judge model or modify its prompts to improve the correlation, with the ultimate goal of achieving a correlation coefficient of 0.75 or higher.

This evaluation framework enables DianGPT with a systematic assessment standard that can accurately quantify the model's performance in multi-turn dialogues, domain knowledge answers, and other aspects. By analyzing the correlation between human scores and the judge model's scores, we obtain a reliable and objective evaluation of the model's performance, offering valuable insights for future model optimization.

TABLE I
EXPERIMENT RESULTS WITH DIFFERENT STRATEGIES WITH DEEPSEEK-V3 AS JUDGE MODEL

Strategy	LLM score	Rouge-L F1	Latency
Base model	3.087	0.137	3.132
SFT model	3.273	0.164	**2.856**
Base model + RAG	**3.587**	**0.242**	6.779
SFT model + RAG	3.329	0.217	6.123

IV. PERFORMANCE EVALUATION

In this section, we conducted a series of tests and evaluations on the DianGPT model, focusing on model performance, fine-tuning results, and the application of RAG techniques. By comparing the performance and efficiency indicated by average inferring latency of each test question of different test models and judge models , we achieved the following results.

A. Model Performance

As shown in Table I, the scoring results of the baseline model (THUDM/glm-4-9b-chat) initially achieved a score of 3.087 for LLM evaluation and 0.137 for Rouge-L F1. After filtering the dataset and fine-tuning the model for 23 hours on an L20 48GB GPU, the performance improved to 3.273 for LLM evaluation and 0.164 for Rouge-L F1. Further enhancement was achieved by incorporating a basic RAG technique, which led to a substantial increase in performance, raising the LLM score to 3.587 and the Rouge-L F1 score to 0.242.

However, the performance would undergo an decline, from 3.587 to 3.329, if the generator of RAG was supervised fine-tuned, resulting from the loss of the model ability of in-context learning as we speculated.

The great improvements mentioned above indicate a significant enhancement in both the accuracy and richness of the generated responses.

Notably, approximately only 18 GB of VRAM is utilized to deploy DianGPT on a single NVIDIA GeForce RTX 3090 under a quantization precision of 4bit, which demonstrates that even with relatively limited computational resources, we were able to achieve a high-performing model, highlighting the efficiency and effectiveness of our optimization strategies.

B. Rationality of the Evaluation Framework

In the judge model evaluation, we selected several different judge models and conducted a correlation analysis. As shown in Table II, the correlation coefficient between the judge model and manual scores was relatively high. Particularly when using the DeepSeek-V3 judge model, the correlation coefficient reached 0.728, as shown in Fig. 2 indicating strong consistency between the judge model's evaluation and the manual scoring. The correlation coefficient of THUDM/glm-4-9b-chat is 0.643, which is satisfactory as well. This comparison further validates the effectiveness and rationality of our evaluation framework in quantifying the model's performance.

TABLE II
CORRELATION COEFFICIENTS OF JUDGE MODELS WITH THE SAME PROMPTS

Judge Model	Correlation Coefficient
Qwen2.5-14B-Instruct	0.669
GLM-4-9b-chat	0.643
DeepSeek-V3	**0.728**

C. Discussions

1) Model Limitations and Challenges: Despite DianGPT's strong performance in domain-specific Q&A, it faces challenges in complex multi-turn dialogues, sometimes exhibiting logical inconsistencies or omissions. Additionally, for long-tail queries, its responses may lack accuracy due to limited exposure to rare domain knowledge. Addressing these issues requires improved contextual understanding and knowledge retrieval mechanisms.

2) Scalability and Practical Applications: DianGPT's scalable framework allows for adaptation across various domains. By integrating domain-specific corpora and knowledge bases, it can be extended to academic fields and industry applications such as customer service and technical support. Further integration with real-world scenarios will enhance its utility and impact.

V. CONCLUSION

In this paper, we presented DianGPT, a domain-specific question-answering system designed to balance precision and efficiency. We detailed its technical framework, which encompasses dataset construction, model fine-tuning via LoRA, RAG-based answer generation, and a comprehensive evaluation framework. Our results demonstrated that combining LoRA-based optimization with RAG notably improves LLM's performance in specialized tasks, even under resource constraints, effectively addressing the two core challenges identified at the outset.

DianGPT exemplifies how tailored fundamental LLMs can be fine-tuned for domain-specific applications, offering a blueprint for adapting intelligent Q&A systems to diverse fields. Beyond its technical contributions, our system demonstrates the potential for innovation in specific domains through strategic resource utilization. Future work will focus on enhancing model robustness and broadening its applicability to areas such as education, collaborative research, and enterprise services, further bridging the gap between domain expertise and accessible LLM-driven applications. We plan some future work as follows:

- **Enhanced Domain Adaptation**: Utilizing domain-adaptive pre-training to improve specialization with minimal labeled data.
- **Multimodal Integration**: Expanding the model's capability to process text, images, and structured data for broader applicability.

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